CODE APEX 3.0 • iMOCHA TRACK

# AI Hackathon Problem Statements

Five problem statements from iMocha, led by Vibe Coding, our AI-era coding assessment.

**Problem statements**

IM-01 Vibe Coding: The AI-Era Coding Assessment
IM-02 Autonomous AI QA Engineer
IM-03 Skill Passport: Verified Skills from Real Work
IM-04 AI Job Simulator: Practise Real Work Scenarios
IM-05 Codebase Mentor: AI Onboarding for Developers

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INTRODUCTION

# About the iMocha track

iMocha is an AI-powered skills intelligence company that helps organisations assess, understand and grow the skills of their people. For CODE APEX 3.0, iMocha is contributing a dedicated track of five problem statements across AI and Generative AI, AI-enabled assessments, skill intelligence, automation, software engineering and QA.

The first problem statement, **Vibe Coding**, comes directly from a product iMocha is prototyping today. The other four focus on real-world problems where AI agents can complete meaningful work and back their results with evidence.

## Problem statements at a glance

| ID | Problem statement | Focus area |
|---|---|---|
| **IM-01** | Vibe Coding: The AI-Era Coding Assessment | AI-enabled assessments • Generative AI |
| **IM-02** | Autonomous AI QA Engineer | QA • Automation • Software engineering |
| **IM-03** | Skill Passport: Verified Skills from Real Work | Skill intelligence • Generative AI |
| **IM-04** | AI Job Simulator: Practise Real Work Scenarios | AI & Generative AI • AI-enabled assessments |
| **IM-05** | Codebase Mentor: AI Onboarding for Developers | Software engineering • Generative AI |

## How each problem statement is structured

- **Background** and **the challenge**: the real problem and why it matters.
- **Expected solution** and **expected outcomes**: what a strong submission should do and show.
- **Stage-wise expectations**: how the same idea should grow across the three stages of the hackathon.

> **One idea, three stages.** Teams choose one problem statement and carry the same idea through Stage 1 (idea and presentation), Stage 2 (online prototype) and Stage 3 (the 24-hour offline grand finale), making it more complete at each stage.

PROBLEM STATEMENT IM-02

# Autonomous AI QA Engineer

| Focus area | Category | Who it is for |
|---|---|---|
| QA • Automation • Software engineering | Software | Developers and QA teams, especially in small teams with little testing capacity |

## Background

Software ships faster than it can be tested. Manual testing is slow and repetitive, automated test suites fall behind new features, and bug reports often arrive without clear steps to reproduce them. Small teams frequently have no dedicated tester at all.

## The challenge

> Build an AI agent that tests a web application the way a careful human tester would: it explores the app, decides what to test, runs the tests, and reports real bugs with evidence.

## Expected solution

- Given a web app address and a short description of what it should do, the agent explores the app like a real user.
- It generates test cases for the main user journeys and edge cases, and runs them in a browser.
- It files clear bug reports: steps to reproduce, expected versus actual behaviour, and screenshots.
- It turns successful journeys into reusable automated tests that the team can keep running.
- Optional: several cooperating agents, for example an explorer, a tester and a reporter.

## Expected outcomes

- On a sample app with deliberately planted bugs, the agent finds and correctly reports a measurable share of them.
- Generated tests run on their own and can be repeated.
- Every bug report can be reproduced by a person following its steps.

## Stage-wise expectations

The same idea is carried through all three stages and grows at each one.

| Stage | What we expect to see |
|---|---|
| **Stage 1**<br>**Idea & presentation** | The testing problem being solved, the agent's design (what it sees, decides and does), the target sample app, and how accuracy and coverage will be measured. |
| **Stage 2**<br>**Online prototype** | The agent tests one key journey (for example sign-up and login) on a sample app, produces a test, and files at least one evidence-backed bug report. |
| **Stage 3**<br>**24-hour grand finale** | Exploration across several journeys, cooperating agents, results on a planted-bug benchmark, and a simple dashboard of findings. |